Project Gaja-Saviya

An Intelligent Multi-Modal Framework for Human-Elephant Conflict Mitigation

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Individual Project Proposal Report

Component: Intelligent Fence Signal Interpreter

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Human-Elephant Conflict (HEC) in Sri Lanka causes extensive crop destruction and danger to farming communities, with existing electric fence systems serving only as crude binary triggers rather than intelligent monitoring tools. These conventional systems detect voltage drops without distinguishing between genuine elephant contact and environmental interference such as falling branches, wind, or equipment degradation. This research proposes the Intelligent Fence Signal Interpreter, an embedded signal processing and machine learning component within Project Gaja-Saviya, designed to transform raw electric fence signals into meaningful, classified disturbance events. The component continuously samples voltage and current waveform patterns from the fence via an ESP32-based Sentinel Node, applies preprocessing techniques including denoising, segmentation, and normalization, and feeds structured time-series sequences into a Long Short-Term Memory (LSTM) neural network. The LSTM model, trained on controlled experimental data representing elephant contact, branch impact, weather noise, and equipment faults, performs multi-class classification of disturbance types. Each classification is accompanied by a confidence score and a precise timestamp. This signal fingerprinting approach replaces simplistic binary voltage threshold detection with a nuanced, intelligent understanding of fence disturbance characteristics, significantly improving the accuracy of upstream threat detection and reducing false alarm propagation through the system.

Keywords: Fence Signal Classification, LSTM, Time-Series Analysis, Electric Fence, IoT, HEC, Signal Fingerprinting

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
HEC	Human-Elephant Conflict
LSTM	Long Short-Term Memory
ML	Machine Learning
ADC	Analog-to-Digital Converter
ESP32	Espressif Systems 32-bit Microcontroller
MQTT	Message Queuing Telemetry Transport
API	Application Programming Interface
RNN	Recurrent Neural Network
FFT	Fast Fourier Transform
SNR	Signal-to-Noise Ratio
RMS	Root Mean Square
DWC	Department of Wildlife Conservation

1. INTRODUCTION

1.1 Background and Context

Electric fences are the most widely deployed physical deterrent used by Sri Lankan farmers to protect their crops from elephant intrusions. These fences deliver high-voltage pulses at regular intervals to discourage animals from crossing. When an elephant contacts the fence, the high impedance of the animal's body causes a characteristic drop in fence voltage. In theory, this voltage drop serves as an intrusion indicator. In practice, however, voltage drops occur constantly due to environmental factors: vegetation growing on fence wires, metal contacts corroding, branches falling across the line, wet weather increasing conductivity, and small animals touching the fence.

The result is that farmers experience frequent false alarms — alerts triggered by events that have nothing to do with elephants. Over time, this leads to alert fatigue, where farmers stop responding to alerts because they assume they are false. A genuine elephant intrusion may then go unnoticed until damage is already done. This is the core problem the Intelligent Fence Signal Interpreter component is designed to solve: replacing crude voltage threshold detection with intelligent signal fingerprinting that can distinguish between these disturbance types.

Electric fence waveforms carry rich information that simple threshold systems ignore. The amplitude, pulse width, rise time, decay profile, and repetition pattern of a disturbance all vary systematically depending on what caused it. An elephant's body produces a distinct impedance signature compared to a wet branch or metal corrosion. By capturing and analyzing these temporal waveform characteristics using machine learning, the system can achieve disturbance-type classification that was previously impossible with conventional electronics.

1.2 Literature Review

Signal classification using machine learning has been extensively studied in electrical fault detection, power grid anomaly detection, and vibration analysis. Early work demonstrated that time-domain features extracted from electrical signals — such as RMS amplitude, zero-crossing rate, and spectral centroid — could distinguish between different fault types using classical classifiers such as SVM and k-NN. However, these approaches require manual feature engineering and do not capture temporal dependencies across signal sequences.

Recurrent Neural Networks, particularly LSTM networks introduced by Hochreiter and Schmidhuber (1997), excel at learning temporal dependencies in sequential data. LSTMs have been successfully applied to power system fault classification, vibration-based anomaly detection in industrial equipment, and biosignal classification. Their ability to maintain long-range dependencies within time-series makes them well-suited to analyzing fence signal sequences, where the pattern of voltage changes over time is more diagnostic than any single instantaneous measurement.

IoT-based fence monitoring systems have been deployed in HEC contexts in South Africa and India. Studies by Kuiper et al. (2018) and Chartier et al. (2020) demonstrated that IoT-connected fence monitors could significantly reduce response times to genuine intrusions. However, these systems still relied on simple voltage threshold detection. The integration of machine learning for disturbance classification in IoT fence monitoring has not been reported in the HEC literature, representing the key technical novelty of this component.

Data preprocessing for electrical signals typically involves denoising using median or Savitzky-Golay filters, segmentation into fixed-length windows, and normalization. Controlled experimental datasets for training such models are constructed by deliberately inducing different disturbance types under monitored conditions — a methodology that this research will adopt.

1.3 Research Gap

The central research gap addressed by this component is the absence of intelligent, multi-class disturbance classification in IoT-based electric fence monitoring systems for HEC. Existing systems treat all voltage anomalies identically, resulting in high false alarm rates that undermine farmer trust. No published system applies LSTM-based time-series classification to electric fence signal data in the context of HEC or wildlife intrusion detection. Furthermore, the concept of signal fingerprinting — learning a characteristic temporal signature for each disturbance type — has not been applied to agricultural electric fences. This research fills this gap by developing, training, and deploying an LSTM-based fence signal classifier as a core component of the Gaja-Saviya framework.

2. OBJECTIVES

2.1 Main Objective

To design and implement an intelligent electric fence signal analysis system that uses LSTM-based time-series classification to identify the type and source of fence disturbances, providing accurate multi-class disturbance labels and confidence scores to the Cross-Modal Sensor Fusion engine.

2.2 Specific Objectives

1. To design and deploy an ESP32-based circuit for continuous sampling of voltage and current waveforms from the electric fence at high temporal resolution.
2. To develop a data collection protocol and conduct controlled experiments to generate labeled training data for four disturbance classes: elephant contact, branch/vegetation impact, weather noise, and equipment fault.
3. To implement a signal preprocessing pipeline including denoising, segmentation into fixed-length windows, and normalization for LSTM input preparation.
4. To design, train, and evaluate an LSTM neural network for multi-class fence disturbance classification.
5. To explore data augmentation techniques for underrepresented disturbance classes to improve model generalization.
6. To define and implement a structured output schema (disturbance type, confidence score, timestamp) compatible with the Cross-Modal Sensor Fusion API.
7. To evaluate model classification performance using precision, recall, F1-score, and confusion matrix, targeting overall accuracy above 88%.

3. METHODOLOGY

This research uses the Design Science Research methodology, building and evaluating a functional LSTM-based fence signal classifier through iterative design and testing cycles.

Phase 1: Hardware Setup and Data Collection

An ESP32-based data acquisition circuit will be designed and connected to an existing electric fence installation at a test site. The circuit will sample fence voltage at 500 Hz using the ESP32 ADC, capturing sufficient temporal resolution to characterize disturbance waveforms. Controlled experiments will be conducted by deliberately inducing each disturbance type — researcher contact simulating elephant impedance using a resistance network, branch placement, water spraying for weather conditions, and deliberate connector corrosion — under observation. Each event will be labeled manually, producing a ground-truth training dataset of at least 500 samples per class.

Phase 2: Signal Preprocessing Pipeline

Raw ADC samples will be passed through a Savitzky-Golay denoising filter to remove high-frequency noise while preserving waveform shape. The signal will then be segmented into overlapping windows of 200 samples (0.4 seconds at 500 Hz) with a 50% overlap. Each window will be normalized using Z-score normalization to remove amplitude-level differences attributable to fence length variation. The resulting normalized time-series windows will be stored as NumPy arrays ready for LSTM input.

Phase 3: LSTM Model Design and Training

An LSTM architecture will be implemented using TensorFlow/Keras. The network will comprise two stacked LSTM layers (64 and 32 units respectively) with dropout regularization, followed by a Dense classification head with softmax activation for four-class output. The model will be trained on the collected dataset using Adam optimizer and categorical cross-entropy loss, with 80/10/10 train/validation/test split. SMOTE oversampling will be applied to address class imbalance for rare disturbance types. Hyperparameter tuning will explore window size, LSTM units, dropout rates, and learning rate.

Phase 4: Deployment and API Integration

The trained LSTM model will be deployed on the backend server using TensorFlow Serving or a Flask API wrapper. Incoming preprocessed windows from the ESP32 node (transmitted via MQTT) will be queued and batch-processed by the inference server. Classification outputs will be serialized into JSON objects (disturbance_type, confidence_score, timestamp) and dispatched to the Cross-Modal Sensor Fusion API via HTTP POST within 1.5 seconds of signal receipt.

4. SYSTEM ARCHITECTURE

4.1 Component Position

The Intelligent Fence Signal Interpreter is the first upstream producer in the Gaja-Saviya pipeline. It acquires raw electrical data directly from the physical fence infrastructure via the ESP32 Sentinel Node and delivers classified disturbance events to the Cross-Modal Sensor Fusion engine.

4.2 Internal Sub-Components

- Signal Acquisition Module: ESP32-based ADC circuit sampling fence voltage at 500 Hz.
- MQTT Publisher: Transmits raw voltage samples from ESP32 to backend server via MQTT broker.
- Preprocessing Engine: Denoising, segmentation, and normalization of received signal windows.
- LSTM Inference Engine: Loads trained model and classifies incoming preprocessed windows.
- Output Serializer: Formats classification output into JSON and dispatches to Fusion API.
- Database Logger: Persists raw signals, preprocessed windows, and classification history.

4.3 Data Flow

Electric Fence -> ESP32 ADC Circuit -> MQTT Broker -> Preprocessing Engine -> LSTM Inference Engine -> Output Serializer -> Fusion API + Database

5. USER REQUIREMENTS

5.1 Stakeholders

- Farmers: Primary beneficiaries of accurate disturbance classification through reduced false alarms.
- Cross-Modal Sensor Fusion Component: Direct consumer of this component's prediction outputs.
- System Administrators: Responsible for monitoring sensor calibration and model performance.
- Hardware Technicians: Responsible for installing and maintaining the ESP32 sensing circuit.

5.2 Functional Requirements

8. The system shall continuously sample electric fence voltage waveforms at 500 Hz via the ESP32 ADC circuit.
9. The system shall transmit raw signal data from the ESP32 to the backend server via MQTT.
10. The system shall apply denoising, segmentation, and normalization to raw signal windows.
11. The system shall classify each signal window into one of four disturbance types: elephant contact, branch/vegetation, weather noise, or equipment fault.
12. The system shall output a structured JSON prediction object with disturbance type, confidence score, and timestamp.
13. The system shall dispatch predictions to the Fusion API within 1.5 seconds of signal receipt.
14. The system shall log all raw signals and predictions to the backend database.
15. The system shall raise a hardware alert if sensor connectivity to the fence is interrupted.

5.3 Non-Functional Requirements

Requirement	Specification
Accuracy	Multi-class classification accuracy above 88% on test set
Latency	Prediction dispatched within 1.5 seconds of signal receipt
Sampling Rate	500 Hz continuous voltage sampling via ADC
Availability	99% uptime for both ESP32 hardware and inference server
Power Efficiency	ESP32 power consumption within 250 mA during active sampling
Maintainability	Model retraining possible without hardware modifications

6. BUDGET AND JUSTIFICATION

Category	Amount (LKR)
ESP32 Microcontrollers & Sensing Circuit	14,500
Electric Fence Test Setup Materials	9,000
Cloud Server (Inference Hosting)	16,000
Software Tools & Libraries	4,500
Field Data Collection & Transport	7,000
Contingency	4,000
TOTAL	**55,000**

Hardware costs are the distinguishing feature of this component budget, as physical ESP32 sensor circuits and fence test setup materials are required that are not needed by software-only components. Cloud hosting supports the always-on inference server. Field collection costs cover transport to elephant conflict zones for controlled experiments.

7. WORK BREAKDOWN STRUCTURE

Phase	Task	Duration
1	Literature Review & Proposal Preparation	Months 1-2
2	Hardware Circuit Design & Assembly	Months 2-3
3	Controlled Experiment Data Collection	Months 3-4
4	Signal Preprocessing Pipeline Development	Months 4-5
5	LSTM Model Design & Training	Months 5-6
6	Model Evaluation & Hyperparameter Tuning	Month 6
7	Deployment & API Integration	Months 6-7
8	System Integration Testing	Months 7-8
9	Documentation & Final Submission	Months 8-9

8. RISK ANALYSIS

Risk	Impact	Mitigation
Insufficient labeled training data for rare disturbance classes	High	Use SMOTE oversampling and controlled lab experiments to generate balanced datasets
Hardware malfunction or ESP32 connectivity failure	High	Deploy redundant ESP32 units; implement watchdog timer and remote health monitoring
LSTM model overfitting on controlled data	Medium	Apply dropout regularization; validate on independently collected field data
Signal noise variation across different fence installations	Medium	Normalize per-installation; retrain on site-specific data if needed
API schema incompatibility with Fusion component	Low	Define and freeze schema early with team; use contract testing

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